OSM - SOG72 & SOG52 & SOG42

HIGHLIGHTS

- **2x Arm Cortex-A78 + 6x Arm Cortex-A55** deliver high performance and power efficiency, built on TSMC 6nm process
- MediaTek 8th generation **NPU with 6.1 - 9 TOPS, total up to 7.2 - 10.5 TOPS** acceleration, enabling **real-time AI**, GenAI applications with **leading LLMs**
- **Arm Mali-G57 MC2 GPU** and advanced multimedia pipelines support up to **4K display and 4K video codecs**

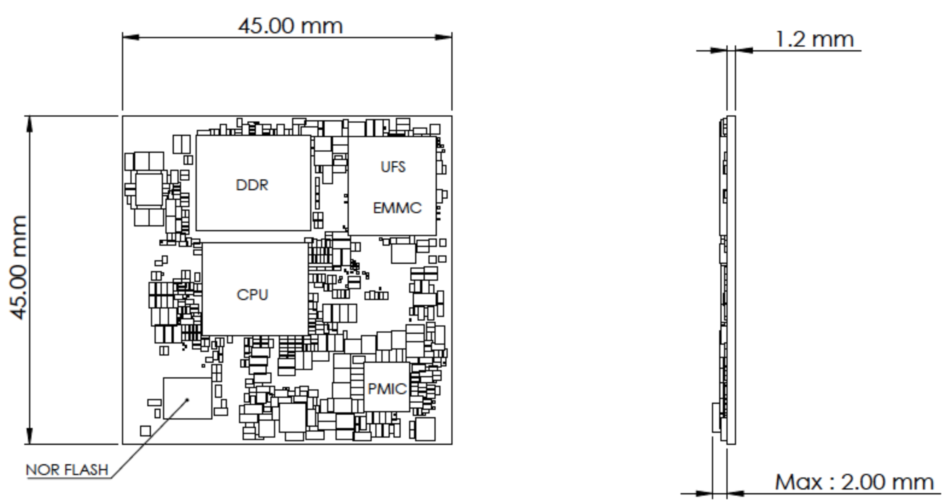
FEATURE

- Power-Efficient Edge GenAI Platform
- Advanced GenAI and Neural Processing
- Advanced Multimedia and Graphics
- Android 16 & Yocto 5.0 support

POTENTIAL APPLICATIONS

- Smart Home
- Industrial
- Smart Retail
- Enterprise
- Healthcare
- Smart Cities

BOARD PLACEMENT



InnoComm Mobile Technology Corporation
3F, No.6, HsinAnn Rd, Hsinchu Science Park, Hsinchu 30078, Taiwan
Phone: +886-3-5781868
Email: innocomm@innocomm.com

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NOTE: This is an early product announcement. All provided information is preliminary and subject to change.

SPECIFICATION

Processor	
CPU (SOG72)	2x Arm Cortex-A78 2.6GHz 6x Arm Cortex-A55 2.0GHz
CPU (SOG52)	2x Arm Cortex-A78 2.2GHz 6x Arm Cortex-A55 2.0GHz
CPU (SOG42)	2x Arm Cortex-A78 1.8GHz 6x Arm Cortex-A55 1.6GHz
GPU (SOG72)	Arm Mali-G57 MC2 1.1GHz OpenGL ES 3.2, Vulkan 1.1, OpenCL 2.2
GPU (SOG52)	Arm Mali-G57 MC2 880MHz OpenGL ES 3.2, Vulkan 1.1, OpenCL 2.2
GPU (SOG42)	Arm Mali-G57 MC2 880MHz OpenGL ES 3.2, Vulkan 1.1, OpenCL 2.2
NPU (SOG72 &SOG52)	8th-Gen NPU 9 TOPS (1x MDLA5.3, GenAI)
NPU (SOG42)	MediaTek 8th-Gen NPU 6.1 TOPS (1x MDLA5.3, GenAI)
Memory and Type	
	LPDDR5(X)-4GB/8GB/16GB
Storage Type	
	UFS3.1 128GB
or	eMMC5.1 32GB/64GB
Display Interface	
	1x MIPI DSI 4-lane
	1x eDP1.4 4-lane
	1x DP1.4/eDP1.2 4-lane (Support DP Alt mode)
Image Signal Processor	
SOG72 & SOG52 & SOG42	Supports up to 6x FHD30 cameras with MIPI virtual channels
SOG 72	2x ISP (3A, NR, AI-FD, LSC, Warp Engine)
	Single camera: 32MP @ 30fps
	Dual camera: 16MP + 16MP @ 30fps
SOG 52 & SOG42	1x ISP (3A, NR, AI-FD, LSC, Warp Engine)
	Single camera: 16MP @ 30fps

Audio	
	2x I2SIN
	2x I2SOUT
Camera	
	2x MIPI CSI 4-lane
Mechanical & Environmental	
Form Factor	45 mm x 45 mm x 3.2 mm
Temp. Range	-20°C to 80°C
Input/Output	
	2x USB 2.0 (host)
	1x USB 3.2 Gen1 (host)
	1x USB 3.2 Gen1(DRD, combo with DP1.4)
	1x Ethernet (10M/100M/1000M)
	3x PWM
	28x GPIOs
	1x PCIe Gen2 1-lane
	1x UART for debug
	3x UARTs for application
	5x I2C
	2x SPI
	2x ADC
	2x SDIO

Note: The difference between SOG72 / SOG52 / SOG42 is limited to CPU / GPU / NPU performance; all other specifications are identical.

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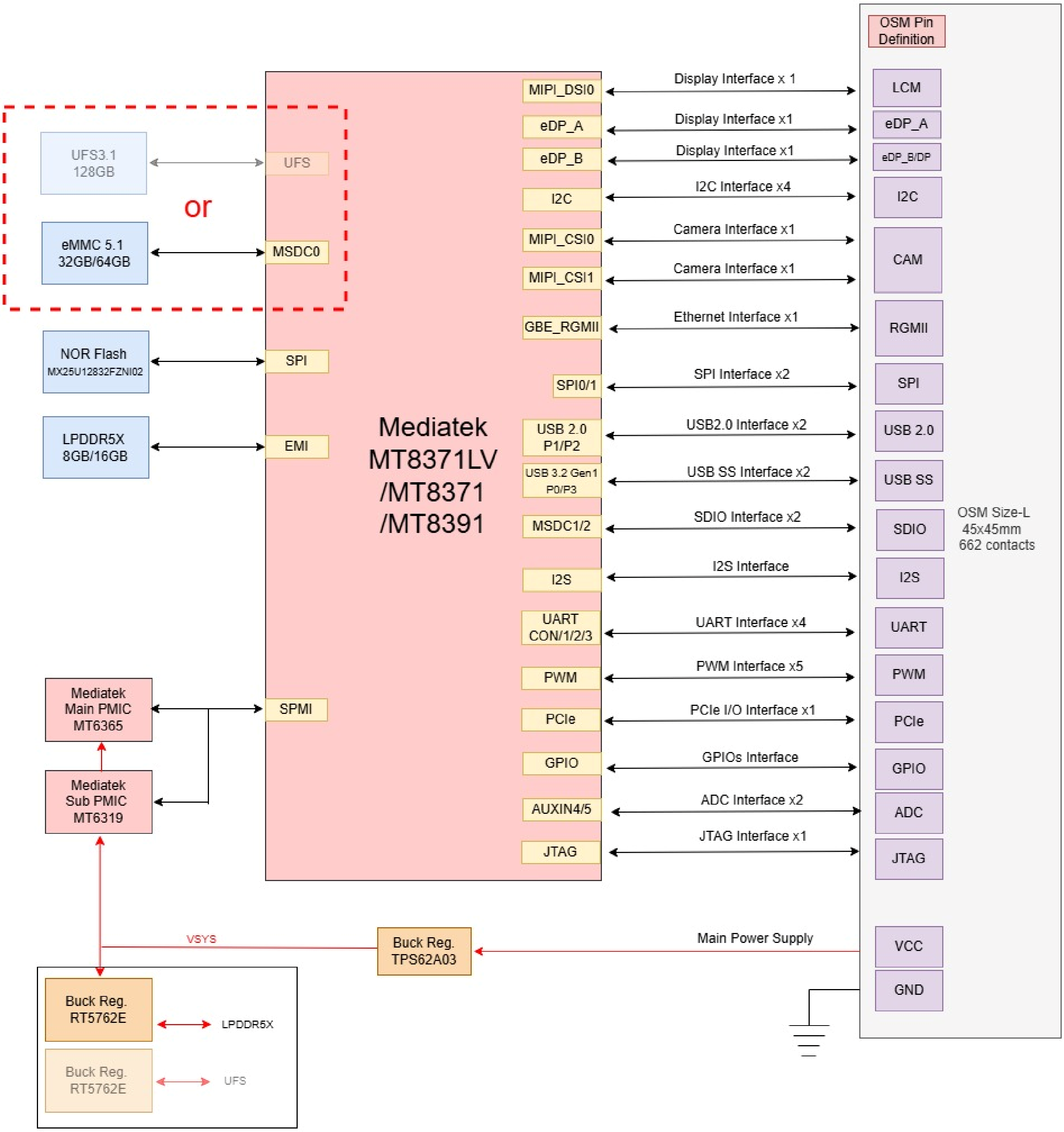
BLOCK DIAGRAM



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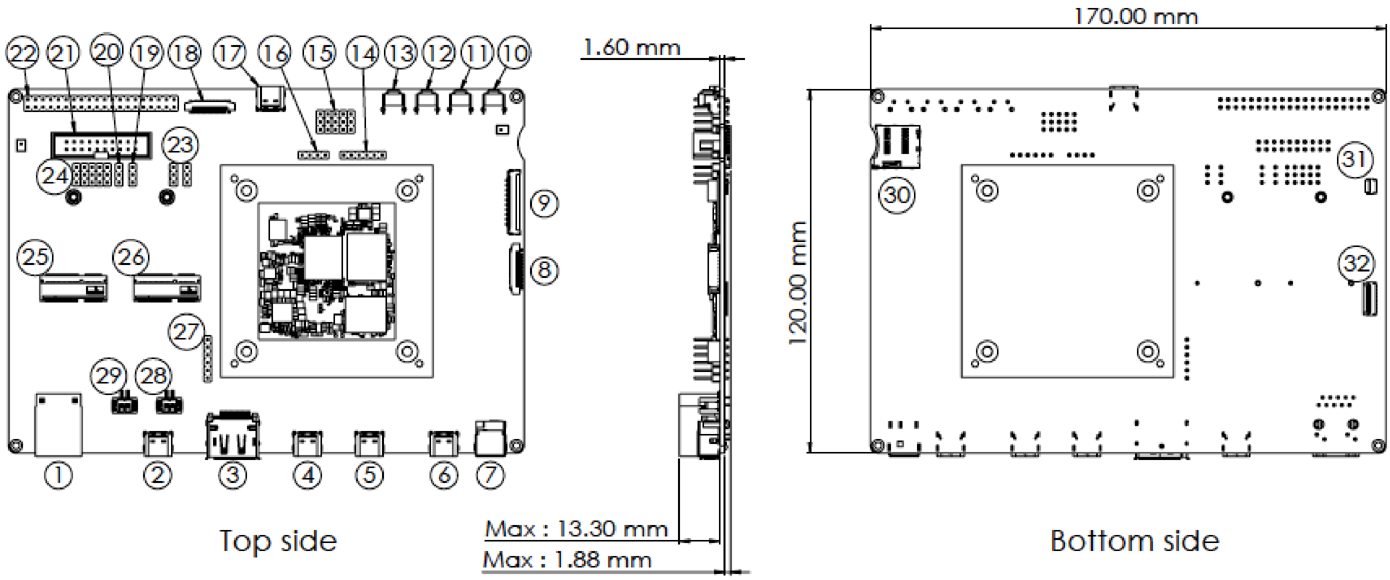
OSM Carrier Board

FEATURE

- Supports **one ultra-wide 4K display** or **dual 2.5K displays**.
- External support for **Wi-Fi 5 (MT-7663)** or **Wi-Fi 6 (MT-7921)** via SDIO or PCIe modules.
- Dual ISP(For SOG72) support: e.g. 2×16MP or a single 32MP image sensor at 30fps.

BOARD PLACEMENT



No.	Function	No.	Function	No.	Function	No.	Function
1	RJ45	11	Reset Key	21	JTAG	31	Touch panel interface
2	USB Type C (D)	12	Vol up/Download	22	Raspberry Pi	32	DSI interface
3	Display Port	13	Vol down	23	Boot SEL		
4	USB Type C (A)	14	SPI_A	24	PCM SEL		
5	UART Debug Port	15	I2S/DMIC	25	M.2 E key PCIe		
6	USB Type C (Power)	16	I2C_A	26	M.2 E key SDIO_B		
7	DC Jack	17	USB Type C (C)	27	I2S_A		
8	CAM _B	18	CAM_A	28	SPK L		
9	eDP	19	RSPI-EN	29	SPK R		
10	Power Key	20	JTAG-EN	30	SD Card		

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OSM Carrier Board

SPECIFICATION

Compatibility	Open Standard Module v1.2
Network	1x RJ45 GbE
Main / Second Display	1x MIPI DSI 4-lane 1x eDP1.4 4-lane DP1.4 4-lane (Display port)
Camera	2x 4-lane MIPI-CSI
Audio	Audio AMP AD82129 (2x SPK connector) or 2x DMIC 1x I2SIN 6PIN connector
USB	1x USB 2.0 Type C (DRD) 1x USB 3.2 Gen1 Type C (host) 1x USB 2.0 Type C (host)
Debug Port	1x USB2.0 Type C interface connector
SPI	1x SPI 6PIN connector (1.8V)
I2C	1x I2C 4PIN connector (1.8V)
Storage	1x microSD slot
RSPi	1x 40PIN RSPi connector
JTAG	1x 20PIN JTAG connector (Shared by GPIOs of RSPi)
Switch & Buttons	1x Power Button 1x Reset Button 1x Vol Up/Download 1x Vol Down
Wireless	2x M.2 2230 key E slot
Power	DC JACK 12V-IN USB-C PD
Dimension	170mm * 120mm
Operating Temperature	0° C to +70 ° C

***Supported display scenario combinations are specified.**

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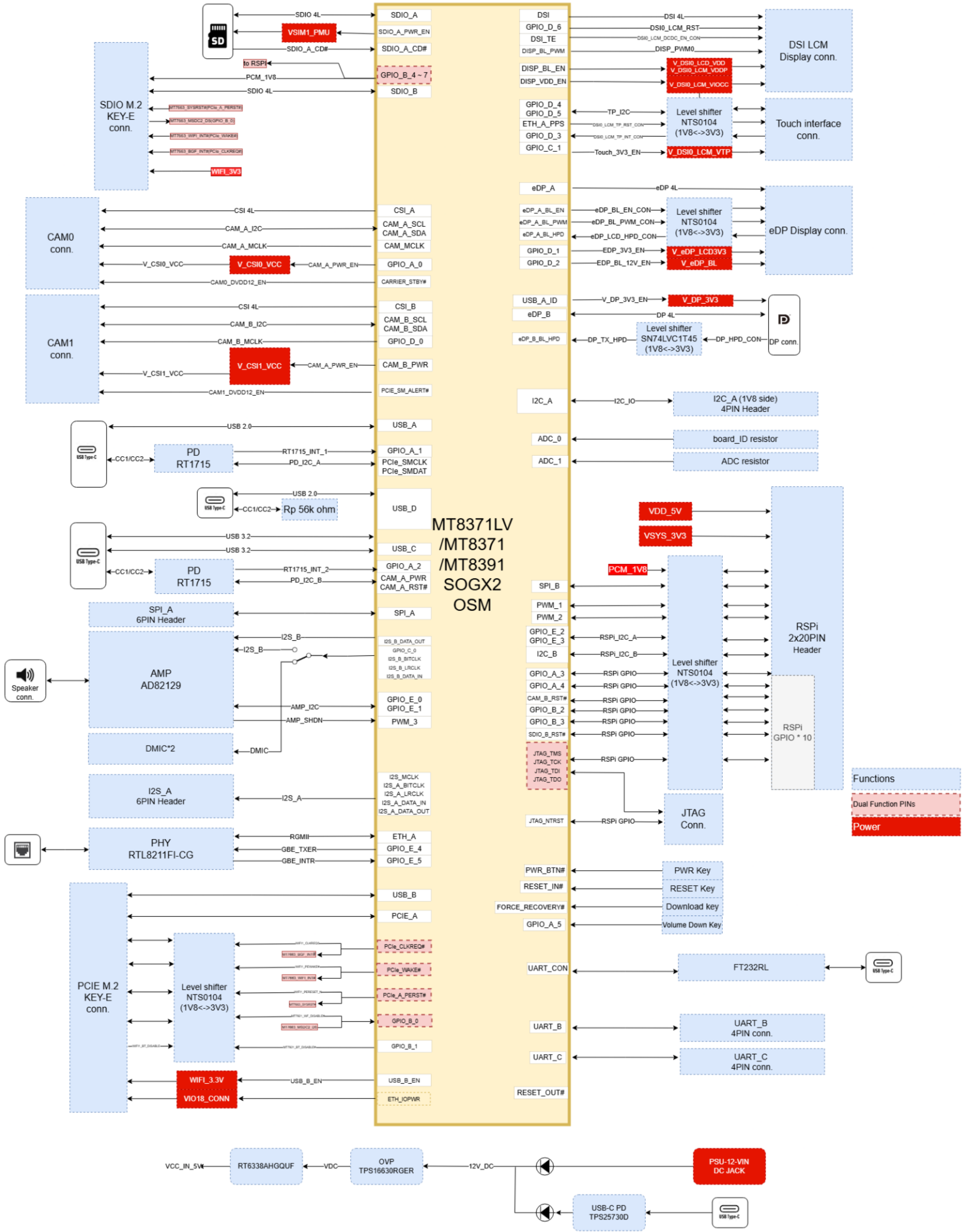

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